

Ex 1.9.

P3. (Use Method I)

$$\text{Here } M(t, y) = 2t \sin y + y^3 e^t$$

$$N(t, y) = t^2 \cos y + 3y^2 e^t$$

This equation is exact since

$$\frac{\partial M}{\partial y} = 2t \cos y + 3y^2 e^t = \frac{\partial N}{\partial t}$$

Hence, there exists a  $\phi(t, y)$

such that

$$(i) \frac{\partial \phi}{\partial t} = 2t \sin y + y^3 e^t$$

$$(ii) \frac{\partial \phi}{\partial y} = t^2 \cos y + 3y^2 e^t$$

From (i), we get

$$\begin{aligned} \phi(t, y) &= \int (2t \sin y + y^3 e^t) dt + h(y) \\ &= t^2 \sin y + y^3 e^t + h(y) \end{aligned}$$

Using (ii), we have

$$t^2 y + 3y^2 e^t = t^2 \cos y + 3y^2 e^t + h'(y)$$

$$\Rightarrow h'(y) = 0 \Rightarrow h(y) = 0$$

So  $\phi(t, y) = t^2 \sin y + y^3 e^t = C$  is the general solution.

P5. (Use Method II)

$$\text{Here } M(t, y) = y \sec^2 t + \sec t \tan t$$

$$N(t, y) = 2y + \tan t$$

This equation is exact since

$$\frac{\partial M}{\partial y} = \sec^2 t = \frac{\partial N}{\partial t}$$

Hence, there exists a  $\phi(t, y)$  such that

$$(i) \frac{\partial \phi}{\partial t} = y \sec^2 t + \sec t \tan t$$

$$(ii) \frac{\partial \phi}{\partial y} = 2y + \tan t$$

From (ii), we obtain

$$\begin{aligned} \phi(t, y) &= \int (2y + \tan t) dy + k(t) \\ &= y^2 + y \tan t + k(t) \end{aligned}$$

Using (i), we get

$$y \sec^2 t + \sec t \tan t = y \sec^2 t + k'(t)$$

$$\Rightarrow k'(t) = \sec t \tan t$$

$$\begin{aligned} \Rightarrow k(t) &= \int \sec t \tan t dt \\ &= \sec t \end{aligned}$$

So  $\phi(t, y) = y^2 + y \tan t + \sec t = C$  is the general solution.

P.6. (Using Integrating factor and Method III)

$$\text{Here } M(t, y) = \frac{y^2}{2} - 2y e^t$$

$$N(t, y) = y - e^t$$

This equation is not exact since

$$\frac{\partial M}{\partial y} = y - 2e^t \neq -e^t = \frac{\partial N}{\partial t}$$

However,

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) = \frac{1}{y - e^t} (y - e^t) = 1$$

Hence, this equation has

$$\mu(t) = e^{\int 1 dt} = e^t \text{ as an integrating factor.}$$

Therefore, there exists a function  $\phi(t, y)$

such that

$$(i) \frac{\partial \phi}{\partial t} = \frac{y^2}{2} e^t - 2y e^{2t}$$

$$(ii) \frac{\partial \phi}{\partial y} = y e^t - e^{2t}$$

From (i) & (ii), we obtain respectively

$$\begin{aligned} \phi(t, y) &= \int \left( \frac{y^2}{2} e^t - 2y e^{2t} \right) dt + h(y) \\ &= \frac{y^2}{2} e^t - y e^{2t} + h(y) \end{aligned}$$

$$\begin{aligned} \phi(t, y) &= \int (y e^t - e^{2t}) dy + k(t) \\ &= \frac{y^2}{2} e^t - y e^{2t} + k(t) \end{aligned}$$

$$\Rightarrow k(t) = h(y) = 0$$

So the general solution is given by

$$\phi(t, y) = \frac{y^2}{2} e^t - y e^{2t} = C$$

P7. Here  $M(t, y) = 2ty^3$   
 $N(t, y) = 3t^2y^2$

It is very easy to find  $\phi(t, y) = t^2y^3$

such that  $\frac{\partial \phi}{\partial t} = M$ ,  $\frac{\partial \phi}{\partial y} = N$ .

So the general solution is given by  
 $t^2y^3 = C$

Putting in  $y(1) = 1$ , we get  $C = 1$ .

Thus, the solution of our initial-value problem is  $t^2y^3 = 1$

P8. Here  $M(t, y) = 2t \cos y + 3t^2y$   
 $N(t, y) = t^3 - t^2 \sin y - y$

This equation is exact since

$$\frac{\partial M}{\partial y} = -2t \sin y + 3t^2 = \frac{\partial N}{\partial t}.$$

So there exists a  $\phi(t, y)$  such that

$$(i) \frac{\partial \phi}{\partial t} = 2t \cos y + 3t^2y$$

$$(ii) \frac{\partial \phi}{\partial y} = t^3 - t^2 \sin y - y$$

From (i) & (ii), we have respectively.

$$\begin{aligned} \phi(t, y) &= \int (2t \cos y + 3t^2y) dt + h(y) \\ &= t^2 \cos y + t^3y + h(y) \end{aligned}$$

$$\begin{aligned} \phi(t, y) &= \int (t^3 - t^2 \sin y - y) dy + k(t) \\ &= t^3y + t^2 \cos y - \frac{y^2}{2} + k(t) \end{aligned}$$

It is easy to find

$$h(y) = -\frac{y^2}{2}, \quad k(t) = 0$$

So the general solution is  
 $\phi(t, y) = t^3y + t^2 \cos y - \frac{y^2}{2} = C$ .

Putting in  $y(0) = 2$ , we get  $C = -2$ .

Thus, the solution of our initial-value problem is  $t^3y + t^2 \cos y - \frac{y^2}{2} = -2$

P11. Here  $M(t, y) = 3ty + y^2$   
 $N(t, y) = t^2 + ty$

This equation is not exact since

$$\frac{\partial M}{\partial y} = 3t + 2y \neq 2t + y = \frac{\partial N}{\partial t}$$

However,

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) = \frac{1}{t^2 + ty} (t + y) = \frac{1}{t}$$

Hence, this equation has

$$M(t) = e^{\int \frac{1}{t} dt} = e^{\ln t} = t$$

as an integrating factor.

Therefore, there exists a  $\phi(t, y)$  s.t.

$$(i) \frac{\partial \phi}{\partial t} = 3t^2y + ty^2$$

$$(ii) \frac{\partial \phi}{\partial y} = t^3 + t^2y$$

From (i) & (ii), we have respectively.

$$\begin{aligned} \phi(t, y) &= \int (3t^2y + ty^2) dt + h(y) \\ &= t^3y + \frac{1}{2}t^2y^2 + h(y) \end{aligned}$$

$$\begin{aligned} \phi(t, y) &= \int (t^3 + t^2y) dy + k(t) \\ &= t^3y + \frac{1}{2}t^2y^2 + k(t) \end{aligned}$$

It is easy to find  $h(y) = k(t) = 0$ .

So the general solution is given by

$$\phi(t, y) = t^3y + \frac{1}{2}t^2y^2 = C$$

Putting in  $y(2) = 1$ , we get

$$C = 2^3 \cdot 1 + \frac{1}{2} \cdot 2^2 \cdot 1^2 = 10$$

So the solution of our initial value problem

is  $t^3y + \frac{1}{2}t^2y^2 = 10$

P12. Here  $M(t, y) = t + ye^{2ty}$

$$N(t, y) = ate^{2ty}$$

So we must have a such that

$$\frac{\partial M}{\partial y} = e^{2ty} + 2tye^{2ty} = ae^{2ty} + 2aty e^{2ty} = \frac{\partial N}{\partial t}$$

$$\Rightarrow a = 1$$

Then we have.

$$(i) M(t, y) = z + ye^{2ty} = \frac{\partial \phi}{\partial t}$$

$$(ii) N(t, y) = te^{2ty} = \frac{\partial \phi}{\partial y}$$

From (i) & (ii), we get respectively .

$$\begin{aligned}\phi(t, y) &= \int (t + ye^{2ty}) dt + h(y) \\ &= \frac{t^2}{2} + \frac{1}{2} e^{2ty} + h(y)\end{aligned}$$

$$\begin{aligned}\phi(t, y) &= \int te^{2ty} dy + k(t) \\ &= \frac{1}{2} e^{2ty} + k(t)\end{aligned}$$

$$\Rightarrow h(y) = 0, \quad k(t) = \frac{t^2}{2}$$

So the general solution is given by .

$$\phi(t, y) = \underline{\underline{\frac{t^2}{2} + \frac{1}{2} e^{2ty} = C}}$$